

Conceptual Design

Esme Basnet

Credit and thank you go to Professor Bruce Irvin from Portland State University for generously providing me with this material.

ER Modeling

Entity Relationship (ER) modeling is useful for conceptual design

Some companies use ER models to document their ontologies, their agreed sets of jargon and concepts about their business.

Others use it as a clean technology-independent mechanism for designing databases.

Multiple types of graphical representation of ER diagrams

Multiple helpful tools, including draw.io which we use in this class.

For this class

Use <https://draw.io/>

Entities

Represent nouns, things that the enterprise cares about.

Entity Set represents the type/class of nouns and an entity represents a specific instance of that type. And often we use the word “entity” interchangeably for both! Sorry!

Something that can be identified independently

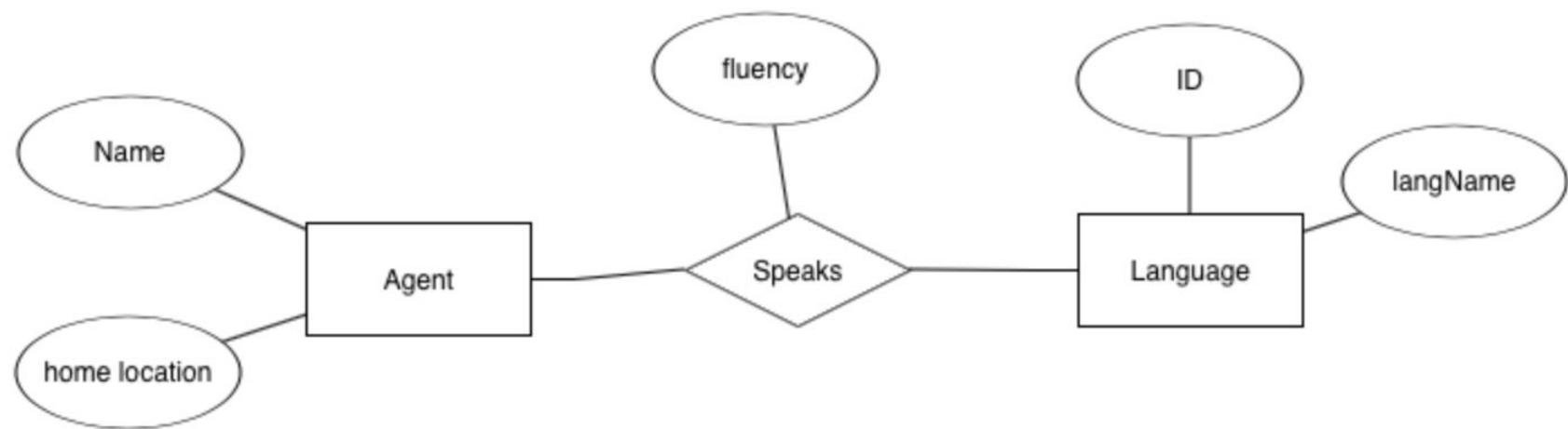
Can be physical or logical

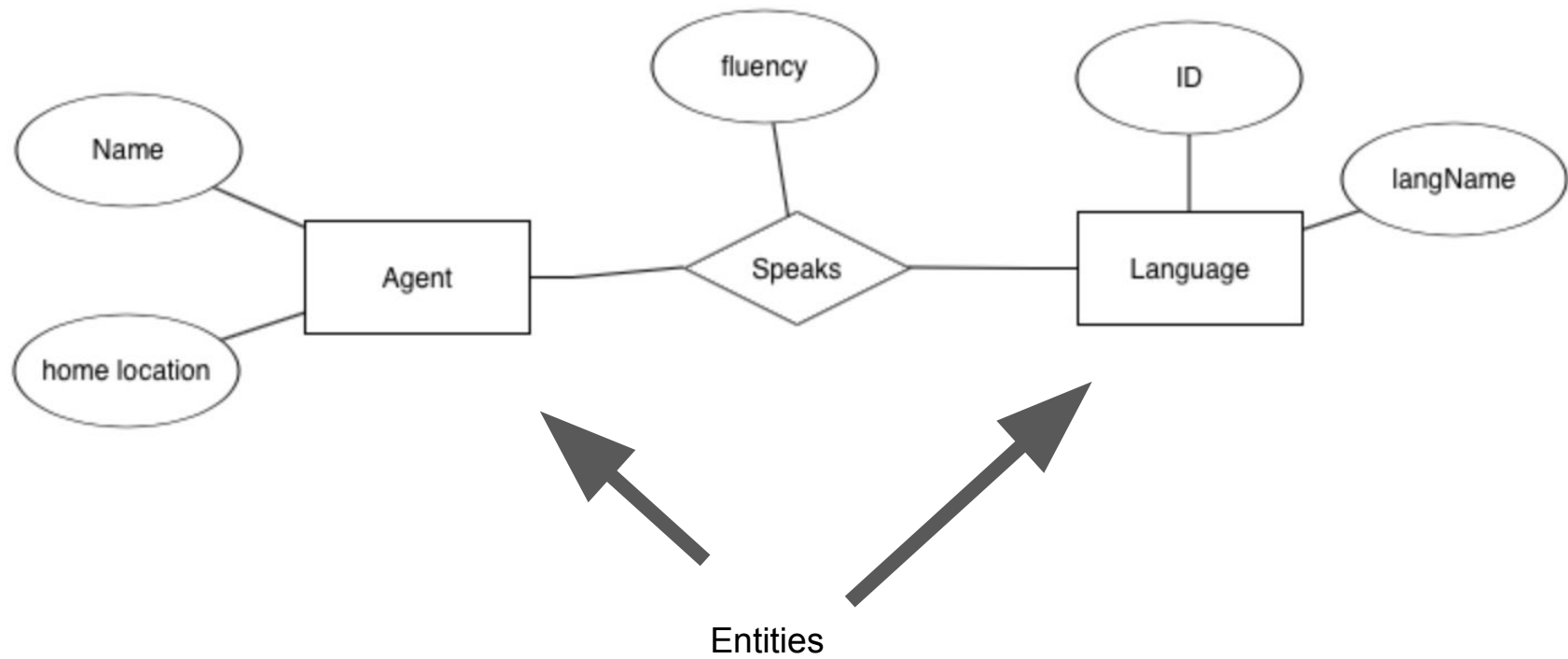
Usually has a primary key

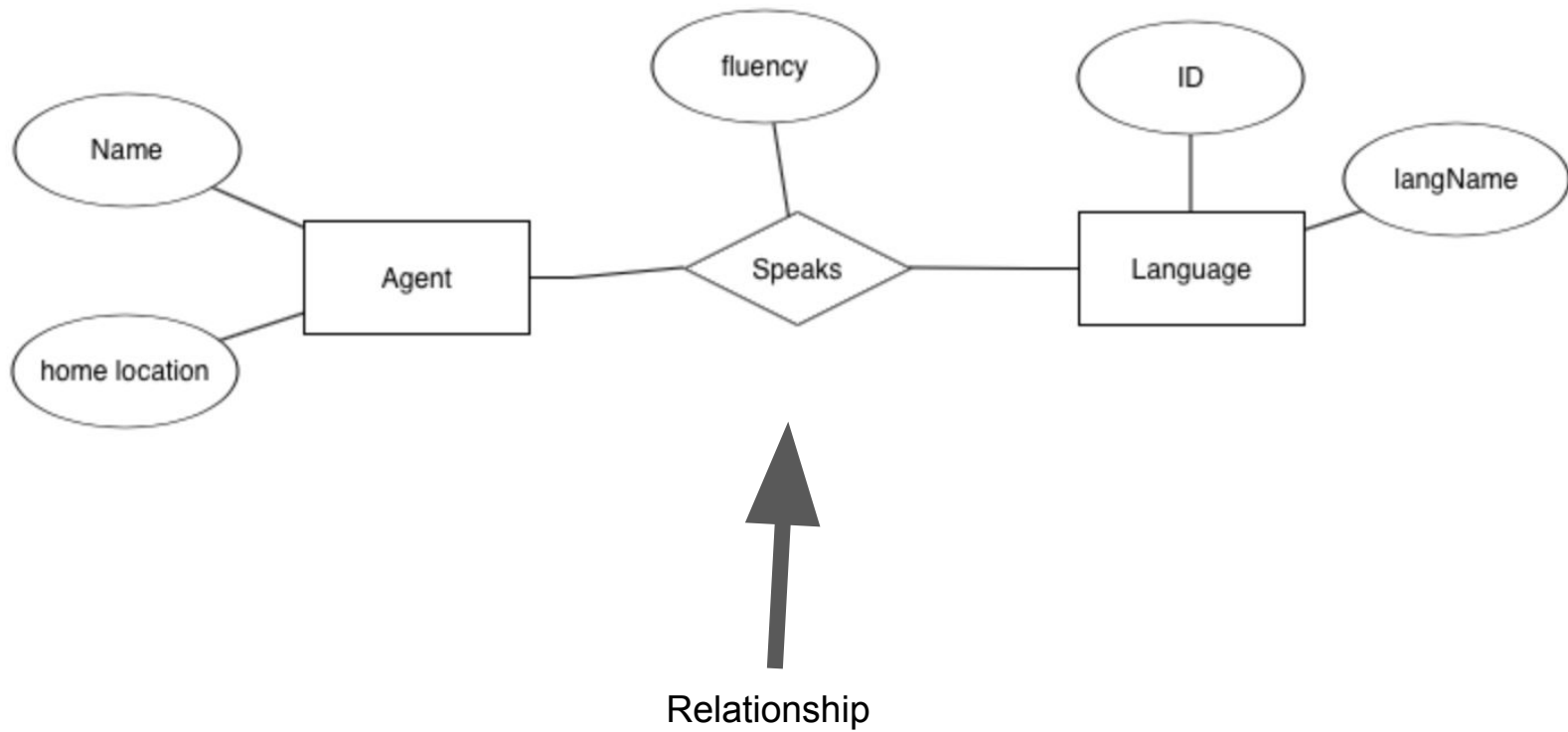
Corresponds to a table in a relational database

Example

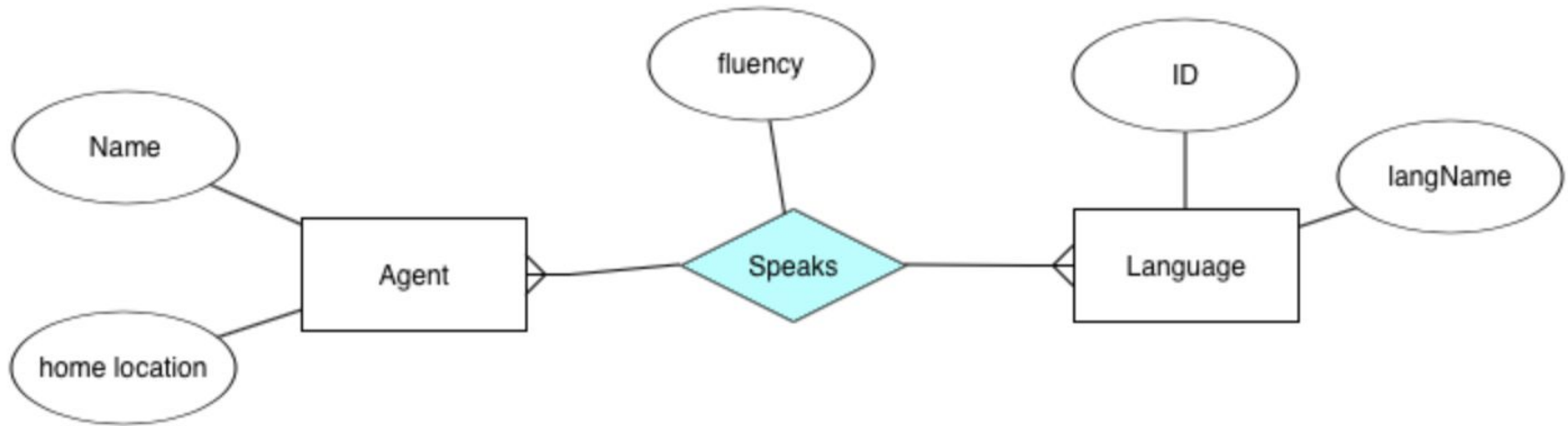
Agents have names and home locations. They speak languages and do so with varying fluency.



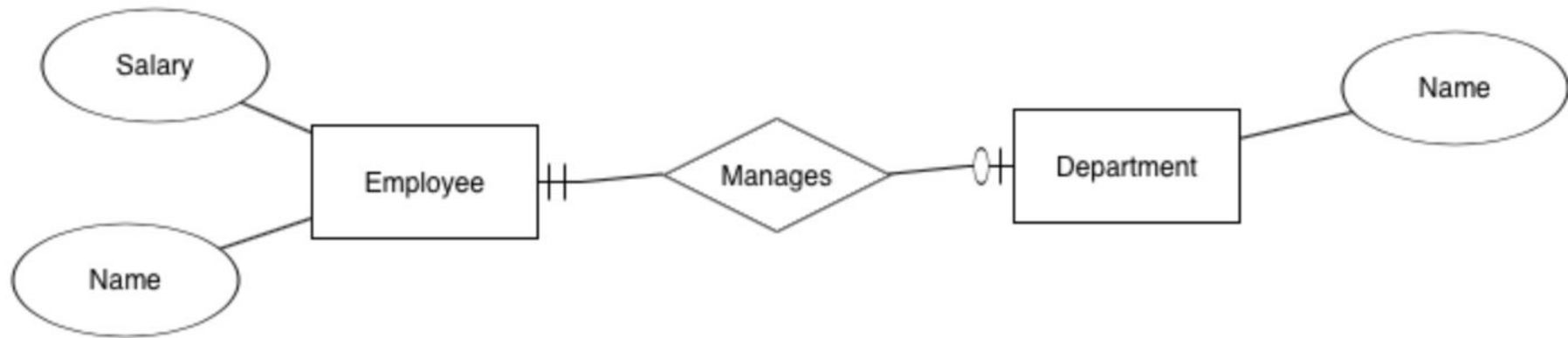




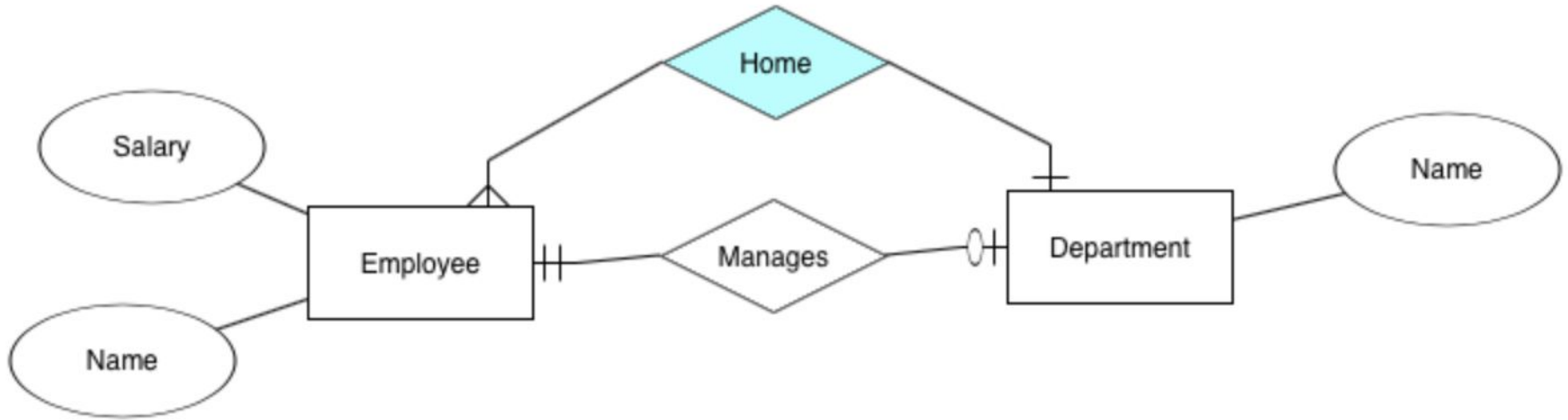
Cardinality: many to many



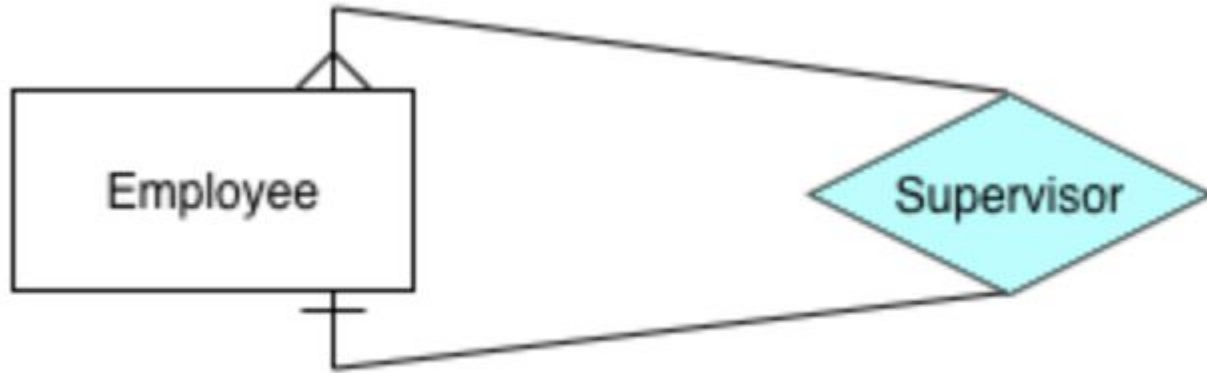
Cardinality: 1 to 1



Cardinality: 1 to Many



Relationships Can be Circular



Weak Entities

WEs are a way for Entity Relationship systems to represent hierarchies, HasA relationships and IsA relationships.

A WE is an entity that only make sense when another related entity exists. I.e., it should be deleted whenever the related entity is deleted.

Might not have it's own primary key (i.e., it's foreign key reference to the related entity helps to form it's primary key)

